TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Identification of Alcohols

How to prove an unknown liquid is an alcohol, sort it into primary, secondary or tertiary with the Lucas timing test, and single out methanol, ethanol, propan-2-ol, tert-butanol and glycerol.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 1 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to identify an unknown alcohol — first to prove that the –OH group is present, then to classify the alcohol as primary (1°), secondary (2°) or tertiary (3°), and finally to confirm individual members by their special tests.

An alcohol is R–OH: a hydroxyl group on a saturated carbon. Three pieces of chemistry do all the identifying. **First**, the O–H hydrogen is *active*: sodium displaces it as hydrogen gas, and ceric ammonium nitrate wraps the oxygen into a red complex — both betray the –OH group. **Second**, substitution of –OH by –Cl under Lucas conditions runs through a carbocation, so the rate follows carbocation stability, 3° > 2° > 1°; the insoluble alkyl chloride appears as cloudiness, and its *timing* reads the class directly. **Third**, oxidation attacks the carbon carrying the –OH only if that carbon still bears a hydrogen: 1° alcohols give aldehydes, 2° give ketones, 3° alcohols resist mild oxidation altogether. The iodoform test is narrower still — it answers only for the CH_3–CH(OH)– unit, or for ethanol, which the reagent oxidises to acetaldehyde (a CH_3–CH=O unit) on the spot.

Preliminary examination first, always: colour and odour (waft, never inhale directly); behaviour on ignition — the lower alcohols burn with a clean, pale-blue, non-smoky flame; reaction to litmus — alcohols are **neutral** (this neutrality, with a positive sodium test, already separates alcohols from acids and phenols, which also react with sodium); and solubility in water — methanol, ethanol and propanols are fully miscible, and miscibility falls steadily as the carbon chain grows.



Figure 1 — Lucas reagent (conc. HCl + anhydrous $ZnCl_2$) converts the alcohol to its chloride; the chloride is insoluble, so the mixture turns cloudy. The waiting time before cloudiness is the classification.

2. Apparatus and reagents

- Dry test tubes, water bath, measuring cylinder, droppers, glass rod, watch glass; the unknown alcohol (about 10 mL).
- Sodium metal (small pieces, stored under kerosene), ceric ammonium nitrate reagent (fresh, yellow), glacial acetic acid and conc. H_2SO_4 (esterification), Lucas reagent (conc. HCl saturated with anhydrous $ZnCl_2$), iodine solution and 10% NaOH (iodoform), acidified $K_2Cr_2O_7$ solution, Schiff's reagent, salicylic acid, potassium hydrogen sulphate ($KHSO_4$) for the glycerol test.

3. Test scheme

Work on 0.5–1 mL portions. Record every observation before moving on — a classification is built from the pattern, never from one test.

TEST	OBSERVATION IF POSITIVE	WHAT IT PROVES / THE CHEMISTRY
Sodium test — a small, dried piece of Na in 1 mL of the *dry* unknown	Brisk effervescence; the gas burns with a pop	Active hydrogen of –OH: 2 ROH + 2 Na → 2 RONa + H_2↑. Acids and phenols also react — neutrality to litmus is what makes this an alcohol's answer.
Ceric ammonium nitrate — 2–3 drops of the yellow reagent	Yellow colour turns **red**	The alcohol oxygen enters a red alkoxy-cerium(IV) complex. (Phenols give a green-to-brown precipitate instead, and are acidic.)

Esterification — warm with glacial acetic acid + 2 drops conc. H_2SO_4, pour into water	Pleasant fruity odour of an ester	$CH_3COOH + ROH \rightleftharpoons CH_3COOR + H_2O$. Confirms a free –OH able to esterify.
Lucas test — 5–6 drops of unknown in 2 mL Lucas reagent, shake, stand, and *time the cloudiness*	Cloudiness from the insoluble alkyl chloride, $ROH + HCl \rightarrow (ZnCl_2)\ RCl + H_2O$	Timing = class; see the decision table in §4. Valid only for alcohols soluble in the reagent (up to about C_6).
Iodoform test — I_2 solution added dropwise with NaOH until pale yellow persists; warm gently	Yellow precipitate of CHI_3 (m.p. 119 °C), antiseptic odour	Needs a CH_3–CH(OH)– unit, or ethanol: $CH_3CH_2OH + 4\ I_2 + 6\ NaOH \rightarrow CHI_3\downarrow + HCOONa + 5\ NaI + 5\ H_2O$. Positive: ethanol, propan-2-ol. Negative: methanol, tert-butanol.
Oxidation — warm with acidified $K_2Cr_2O_7$	1° and 2°: orange turns green (aldehyde, then acid; or ketone). 3°: stays orange	Oxidation needs a hydrogen on the carbinol carbon; a 3° carbon has none and resists mild oxidation.

Confirming individuals. Methanol: oxidise a portion (acidified dichromate, or pass the vapour over a hot copper spiral) to formaldehyde; Schiff's reagent then restores a pink-violet colour quickly. Alternatively warm with salicylic acid and a drop of conc. H_2SO_4 — the oil-of-wintergreen odour of methyl salicylate is unmistakable. Methanol fails the iodoform test: it has no CH_3–CH(OH)– unit. **Ethanol:** the only primary alcohol answering the iodoform test. **Propan-2-ol:** iodoform positive (the reagent oxidises it to acetone first), Lucas cloudiness in about 5 minutes. **tert-Butanol:** Lucas cloudiness at once, no iodoform, resists oxidation. **Glycerol:** a viscous, water-miscible liquid; heated with $KHSO_4$ it gives the sharp, pungent odour of acrolein — $CH_2OH–CHOH–CH_2OH \rightarrow CH_2=CH–CHO + 2\ H_2O$.

4. Summary decision table

TEST	PRIMARY (1°)	SECONDARY (2°)	TERTIARY (3°)
Lucas timing	No cloudiness at room temperature (reacts only on heating)	Cloudiness in about 5 min	Cloudiness almost at once
Acidified $K_2Cr_2O_7$, warm	Orange → green (aldehyde → acid)	Orange → green (ketone)	Unchanged (orange)

Iodoform	Only ethanol positive	Positive if a CH_3–CH(OH)– unit (e.g. propan-2-ol)	Negative
Sodium / CAN / esterification	Positive for all three classes — these prove the –OH group, not the class		

5. Model identification record

WORKED EXAMPLE — UNKNOWN A

Colourless liquid, neutral to litmus, completely miscible with water.
Sodium test: brisk effervescence of H_2. CAN test: red colour. Esterification: fruity odour. → An **alcohol** is confirmed three times over.
Lucas test: mixture still clear after 10 minutes at room temperature. → **Primary** alcohol.
Iodoform test: yellow precipitate of CHI_3. → Of all primary alcohols only **ethanol** answers this test (it is oxidised to acetaldehyde in the reagent).
Conclusion: unknown A is ethanol.

6. Precautions

- The sodium test demands a **dry** sample and dry glassware: water also liberates H_2 from sodium and would fake a positive. Use a rice-grain piece of sodium, dried on filter paper; never return unused sodium to the bench.
- Lucas reagent is concentrated HCl with $ZnCl_2$ — corrosive; make it fresh, and start the timing the moment the alcohol is added, since the whole test is the timing.
- Do not run the Lucas test on an alcohol that does not dissolve in the reagent (long-chain alcohols): an undissolved layer is not cloudiness.
- Smell all unknowns by wafting. Methanol is seriously toxic — small quantities only, no tasting, ever, of any unknown.
- In the iodoform test keep the final mixture just pale yellow: a large excess of iodine darkens the mixture and hides the yellow precipitate.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why does the Lucas test separate the three classes of alcohol when the sodium test cannot?
2. Methanol fails the iodoform test but ethanol passes it. Explain, from the structures.

3. An unknown C_7 primary alcohol gives no Lucas cloudiness even on long standing. Is it therefore not an alcohol? Argue carefully.
4. Phenols also give a colour with ceric ammonium nitrate. How do you keep a phenol from being reported as an alcohol?
5. Why must the sample be dry for the sodium test, and what false conclusion follows if it is wet?
6. Excess ethanol is treated with 0.46 g of sodium. What volume of hydrogen is evolved at STP?

Answer key

1. Lucas substitution goes through a carbocation (S_N1), whose stability runs 3° > 2° > 1°; the rate — read as time to cloudiness — therefore reports the class. The sodium test only detects the O–H hydrogen, which all three classes possess equally.
2. Iodoform needs a CH_3–CO– unit, or a CH_3–CH(OH)– unit the reagent can oxidise to one. Ethanol, CH_3–CH_2OH, is oxidised in the reagent to acetaldehyde, CH_3–CHO, which qualifies. Methanol, CH_3OH, oxidises to formaldehyde, HCHO — no CH_3–CO– unit can form, so no CHI_3.
3. Not necessarily. The Lucas reagent dissolves only the smaller alcohols (up to about C_6). A C_7 alcohol may simply be undissolved, and "no cloudiness" then means "no contact", not "primary". Confirm –OH by sodium/CAN/esterification tests and classify by oxidation behaviour instead.
4. Check the reaction to litmus and the sodium test together: a phenol is acidic and turns blue litmus red, and gives a green-brown precipitate with CAN, whereas an alcohol is neutral and gives a clear red CAN colour.
5. Water reacts with sodium exactly as an alcohol does ($2\ H_2O + 2\ Na \rightarrow 2\ NaOH + H_2$), so a wet sample gives a brisk false positive and an inactive liquid could be misread as an alcohol.
6. 0.46 g Na = 0.46/23 = 0.020 mol. From $2\ Na \rightarrow H_2$, hydrogen formed = 0.010 mol = 0.010 × 22,400 = **224 mL at STP**.